

Counting and the complement rule

Count outcomes without listing them, then turn the counts into probabilities.

THE QUESTION

How many outcomes are there when the list is too long to write, and what is the chance of "at least one"?

THE INTUITION

Multiply the options at each step. When only the group matters, divide by the $k!$ orders of each group. For "at least one", count the opposite case, "none", and subtract.

KEY TAKEAWAY

Ask whether order matters and whether repeats are allowed; that picks the formula. Then $P(\text{at least one}) = 1 - P(\text{none})$.

a sequence of steps → multiply → order matters? → repeats? → $|E| / |S|$ → or $1 - P(\text{none})$

1 Multiply the options at each step



If step 1 has n_1 options and each choice leaves n_2 options for step 2, there are $n_1 \times n_2$ outcomes, one per leaf of the tree; each further step adds a factor. The options at step 2 may differ from branch to branch; only their number must stay the same. Add instead when the cases are disjoint: a code of 3 or 4 digits has $10^3 + 10^4 = 11,000$ options.

2 Permutations: order matters, no repeats

put all 4 of 1, 2, 3, 4 in order: slots 1st to 4th

$$4 \times 3 \times 2 \times 1 = 24 = 4!$$

gold, then silver, among 4 runners

$$4 \times 3 = 12 = 4! / 2!$$

n	n!
0	1
3	6
4	24
5	120
8	40,320
10	3,628,800

Fill k slots from n distinct items: n choices, then $n-1$, down to $n-k+1$. That gives $n!/(n-k)!$ arrangements, and $n!$ orders of all n items (with $0! = 1$). If repeats are allowed, every slot keeps all n options: n^k . `math.perm(4, 2)` is 12; `itertools.permutations(x, 2)` lists them.

3 Combinations: divide out the orders

12 ordered pairs from 1, 2, 3, 4 form 6 groups of $2! = 2$

{1, 2}	{1, 3}	{1, 4}	{2, 3}	{2, 4}	{3, 4}
(1, 2)	(1, 3)	(1, 4)	(2, 3)	(2, 4)	(3, 4)
(2, 1)	(3, 1)	(4, 1)	(3, 2)	(4, 2)	(4, 3)

$k = 3$: the group {1, 2, 3} has $3! = 6$ orders

123 132 213 231 312 321

24 ordered triples / $3! = 4$ groups = $C(4, 3)$

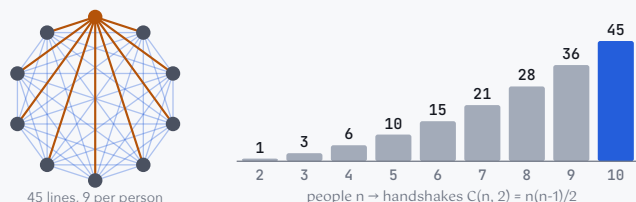
When only the group matters, each group of k appears $k!$ times among the ordered picks, so $\binom{n}{k} = n!/(k!(n-k)!)$, "n choose k". For $x = [1, 2, 3, 4]$, `combinations(x, 2)` gives 6 pairs, and `math.comb(4, 2)` is 6. Choosing who is in fixes who is out: $\binom{n}{k} = \binom{n}{n-k}$.

4 Two questions pick the formula

ordered repeats allowed	ordered no repeats	unordered repeats allowed	unordered no repeats
16 n^k	12 $n! / (n-k)!$	10 $C(n+k-1, k)$	6 $C(n, k)$

Does order matter, and may an item repeat? Here $n = 4$, $k = 2$; cell 23 means 2, then 3. In `itertools`, left to right: `product(x, repeat=2)`, `permutations`, `combinations_with_replacement`, `combinations`. Trap: unordered outcomes with repeats are not equally likely. With two dice, {1, 2} covers two ordered pairs and {1, 1} one, so compute probabilities on the n^k ordered outcomes.

5 Handshakes: 10 people make 45



Each of 10 people shakes 9 hands, but $10 \times 9 = 90$ counts each handshake twice (A with B, B with A): $90/2 = 45 = \binom{10}{2}$. If each person instead sends a card to every other, order matters (A to B differs from B to A): 90 cards. The pairs in a group, $\binom{n}{2} = n(n-1)/2$, grow like n^2 : 23 people make 253 pairs.

6 Pascal's triangle holds every $\binom{n}{k}$

$C(n, k)$: row n , entry k (both from 0)	sum
1	1
1 1	2
1 2 1	4
1 3 3 1	8
1 4 6 4 1	16
1 5 10 10 5 1	32
1 6 15 20 15 6 1	64

$$3 + 3 = 6$$

each entry adds the two above it

$$1 + 4 + 6 + 4 + 1$$

ways to get 0 to 4 heads in 4 coin flips

$$\text{row sums: } 2^n$$

Row n lists $\binom{n}{0}, \dots, \binom{n}{n}$. Item n is in the group or out, so $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. Row n sums to 2^n ; row 6 counts the 64 events of a die (see the sets sheet). Binomial theorem: $(a+b)^n = \sum_k \binom{n}{k} a^k b^{n-k}$.

From counts to probabilities

Count the event and the sample space the same way. For "at least one", count the opposite case.

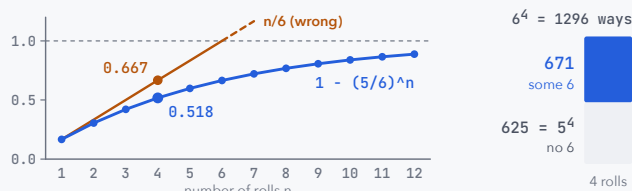
7 Count top and bottom the same way



unordered: 21 pairs, 6 both red
 $C(4, 2) / C(7, 2) = 6/21 = 2/7$
 ordered: 42 pairs, 12 both red
 $(4 \times 3) / (7 \times 6) = 12/42 = 2/7$
 mixed: unordered over ordered
 $6/42 = 1/7$ wrong

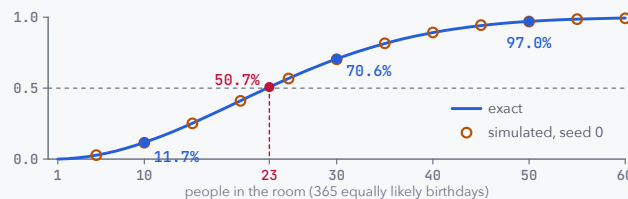
With equally likely outcomes, $P(E) = |E| / |S|$; count both with the same kind of outcome. Draw 2 of 4 red and 3 blue balls, without replacement: $P(\text{both red})$ is $6/21$ as groups and $12/42$ as orders, $2/7$ either way. Mixing them gives $6/42$, off by a factor of $2!$. With replacement, $(4/7)^2 = 16/49$, a little above $2/7 = 14/49$.

8 At least one: subtract "none" from 1



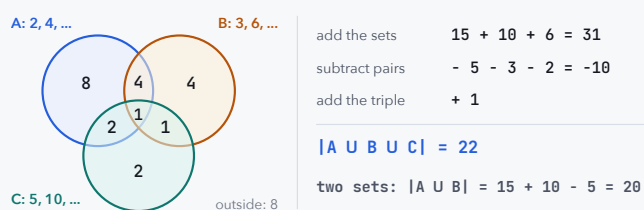
"At least one six in 4 rolls" covers many cases; "no six" is one: $5^4 = 625$ of the $6^4 = 1296$ equally likely sequences. So $P = 671/1296 \approx 0.518$. Adding $1/6$ four times gives 0.667: sequences with two or more sixes are counted more than once, and $n/6$ reaches 1 at 6 rolls. Inclusion-exclusion fixes the count: $4 \times 6^3 - 6 \times 6^2 + 4 \times 6 - 1 = 671$.

9 Birthdays: 23 people pass 50%



All n birthdays differ with probability $365/365 \times 364/365 \times \dots \times (365 - n + 1)/365$; $P(\text{shared})$ is 1 minus that. It passes one half at $n = 23$ (50.7%); 10, 30 and 50 people give 11.7%, 70.6% and 97.0%. It climbs early because 23 people form 253 pairs, and any pair can match. Assumes 365 equally likely days, no 29 February.

10 Inclusion-exclusion removes double counts



Two sets: $|A \cup B| = |A| + |B| - |A \cap B|$; divide by $|S|$ for probabilities. Three sets: add the sets, subtract the pairwise overlaps, add back the triple. In 1 to 30, 15 numbers are multiples of 2, 10 of 3 and 6 of 5; 22 are multiples of at least one, so $30 - 22 = 8$ are multiples of none.

11 List and count in Python, then simulate to check

```
from itertools import (product,
                       permutations, combinations)
from math import comb, perm, factorial
x = [1, 2, 3, 4]
len(list(product(x, repeat=2))) # 16
len(list(permutations(x, 2))) # 12
len(list(combinations(x, 2))) # 6
perm(4, 2), comb(4, 2), comb(10, 2)
# (12, 6, 45): no list needed
```

```
# exact: list all 6^4 sequences of 4 rolls
S = list(product(range(1, 7), repeat=4))
E = [s for s in S if 6 in s]
Fraction(len(E), len(S)) # 671/1296
1 - Fraction(5, 6)**4 # 671/1296
# simulated: 10,000 sets of 4 rolls
rng = np.random.default_rng(0)
r = rng.integers(1, 7, size=(10_000, 4))
(r == 6).any(axis=1).mean() # 0.517
```

12 Pause and think

A committee of 3 from 8 people?
 $\binom{8}{3} = 56$. With a chair, secretary and treasurer: $8 \times 7 \times 6 = 336 = 56 \times 3!$.
 Two dice have 21 unordered outcomes. Is each $1/21$? No: $\{1, 2\}$ has 2 of the 36 ordered pairs, $\{1, 1\}$ has 1.

$P(\text{at least one head in 5 flips})$? Only TTTT misses: $1 - 1/32 = 31/32$.

Why is $\binom{n}{k} = \binom{n}{n-k}$? Picking the k who are in also picks the $n - k$ who are out.

23 people: $P(\text{someone shares your birthday})$? $1 - (364/365)^{22} \approx 5.9\%$: only 22 pairs include you. Disjoint A, B with $P(A) = P(B) = 1/2$. $P(A \cup B)$? 1: the overlap is empty, so nothing is subtracted.

multiply the steps · order matters: $n! / (n-k)!$ · groups: divide by $k!$ · at least one = $1 - P(\text{none})$ · subtract overlaps

Source: the course notebooks "Probability" (itertools combinations, permutations and product; $\binom{4}{2} = 6$ pairs from $[1, 2, 3, 4]$; the event space of a die) and "Set" (complement, union), nipunbatra.github.io/psdv-teaching. The other examples and all numbers are our own, computed by the sheet's figure script; simulations use seed 0. Next: the conditional probability sheet.